

LESSON 5-2 • SESSION 2 • PRACTICE • FORM B

Missing Measures in Triangles

Learning target – I can find a missing base or height of a triangle when I know its area. • 6.G.A.1

REMEMBER

- Work backward from $A = \frac{1}{2} \times b \times h$.
- **Double the area** to undo the half, then **divide** by the measure you know.
- Check: $\frac{1}{2} \times b \times h$ gives the area. A base or height is in cm, not cm^2 .

WORKED EXAMPLEArea 40 cm^2 , base 10 cm . Find the height.

Step 1 $40 = \frac{1}{2} \times 10 \times h$

Step 2 $2 \times 40 = 80$

Step 3 $80 \div 10 = 8$

Answer 8 cm **WORDS** area (*área*) • formula (*fórmula*) • inverse operation (*operación inversa*)**PRACTICE**

show your work • label units

1. Find the missing measure of each triangle.

a. $A = 24 \text{ cm}^2$, $h = 6 \text{ cm}$; $b = \underline{\hspace{2cm}}$

b. $A = 40 \text{ in}^2$, $b = 10 \text{ in}$; $h = \underline{\hspace{2cm}}$

c. $A = 35 \text{ m}^2$, $h = 7 \text{ m}$; $b = \underline{\hspace{2cm}}$

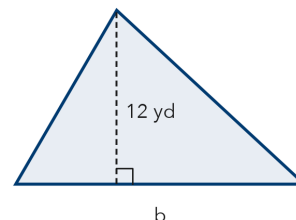
d. $A = 27 \text{ ft}^2$, $b = 9 \text{ ft}$; $h = \underline{\hspace{2cm}}$

- 2.
- Park Lot.**
- A triangular lot covers
- 132 yd^2
- . Its height is
- 12 yd
- .

Hint Double the area first.

a. Fill in the formula: $132 = \frac{1}{2} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$

b. Find the base b . $\underline{\hspace{2cm}} \text{ yd}$



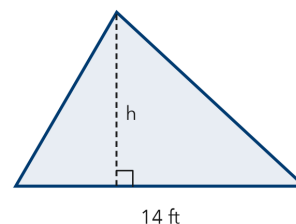
- 3.
- Quilt Piece.**
- A triangular quilt piece has an area of
- 42 in^2
- and a base of
- 7 in
- . Find the height.

Answer _____ in.

- 4.
- Roof Gable.**
- A triangular gable covers
- 135 ft^2
- . Its height is
- 15 ft
- . Find the base.

Answer _____ ft

- 5.
- Awning Panel.**
- The awning panel covers
- 70 ft^2
- . Find the height
- h
- .

Hint Undo the $\frac{1}{2}$ by doubling 70.**Answer** _____ ft

- 6. Banner Claim.** A triangular banner has an area of 90 ft^2 and a height of 9 ft. Deshawn says the base must be 10 ft. Is he right? Use the formula.

Hint Put Deshawn's base into $\frac{1}{2} \times b \times h$ and check.

Show your work.

Base _____

- 7. Sandbox Wedge.** A triangular wedge of a sandbox covers 52 m^2 . Its height is 8 m. Find the base.

Show your work.

Answer _____ m

- 8. Two Signs.** Sign A is a triangle with area 150 ft^2 and base 20 ft. Sign B is a triangle with area 150 ft^2 and height 10 ft.

- Find the height of Sign A. _____ ft
- Find the base of Sign B. _____ ft
- Both signs have the same area. How can their shapes be different?

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Ravi's Work.** A triangle has an area of 90 cm^2 and a base of 15 cm. Ravi wrote: " $90 \div 15 = 6$, so the height is 6 cm." Explain his mistake. Then find the height.

TEST PRACTICE

choose the best answer

- 10.** A triangle has area 48 m^2 and base 12 m. Its height is –

A 4 m **B** 8 m **C** 36 m **D** 576 m

- 11.** A triangle has area 100 in^2 and height 20 in. Its base is –

A 5 in. **B** 80 in. **C** 10 in. **D** 2,000 in.

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

30 E	18 V	10 E	9 Z	20 R	5 X	22 I	13 S	11 Q	12 N
#2b		#3	#4	#5	#6	#7	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____